

MANTHAN SHRIRAM JOSHI

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Education

Northeastern University, Boston MA Master of Science in Cell and Gene Therapies Coursework: Cell and Gene Therapies, Advanced Drug Delivery Systems, Regulatory Landscapes, Immunology, Biotech Enterprise	December 2026
Harvard Business School, Boston MA Certificate in Management Essentials	August 2025
Manipal Institute of Technology (MAHE), Manipal India Bachelor of Technology in Biotechnology Minor: Pharmaceutical Biotechnology Four-year accredited engineering program. Coursework: Bioprocess Engineering, Downstream Processing, Bioreaction Engineering	July 2023

Skills

Process Development	Process characterization and criticality assessment (QTPP, CQA, CPP); deviation investigation, root-cause analysis and CAPA; in-process control and acceptance criteria definition; assay system suitability (blank, full-kill control, Z'); process optimization and yield improvement; protocol and SOP authorship; technical report writing.
Cell Culture & Editing	Aseptic technique; mammalian cell culture (CHO) across 5+ passages at >80% viability; expansion, banking and cryopreservation; end-to-end CRISPR/Cas9 knock-out and knock-in workflow design; Cas9-RNP assembly with target-specific sgRNAs; CRISPRmax lipofection; cytotoxicity and kill-curve assay design (CellTiter-Glo 2.0).
Molecular & Analytical	Genomic DNA extraction; Phusion PCR; primer design in Benchling; agarose gel electrophoresis; amplicon clean-up and Nanodrop QC; Sanger sequencing and ICE indel deconvolution; restriction-enzyme digestion for construct verification; bacterial transformation, colony screening and miniprep; recombinant protein expression and Ni-NTA purification (IMAC); SDS-PAGE; Western blotting; ELISA.
CGT Domain Knowledge	CAR-T (autologous and allogeneic platforms); iPSC differentiation; lentiviral vector handling and titer determination; cGMP and CGT regulatory frameworks (coursework).
Commercial & Business	Market research and competitive landscaping; investor deck development; grant proposal support; client and stakeholder communication.
Instrumentation & Software	Bioreactor/fermenter operation; HPLC; gas chromatography; fluorescence microscopy; flow cytometry sample preparation; FlowJo; Benchling (primer design, construct mapping); GraphPad Prism (dose-response curve fitting); ICE web tool; Microsoft Excel and Office; R (basic).

Professional Experience

Northeastern University, Teaching Assistant, Cell and Gene Therapies, Boston, MA • Taught graduate-level CGT concepts in office hours and discussion sessions, translating current CAR-T and iPSC-to-T-cell literature for students from mixed scientific backgrounds. • Reviewed student project deliverables and delivered structured written feedback; prepared literature summaries on allogeneic and autologous CAR-T platforms to support classroom case discussions.	Jan 2026 – Apr 2026
Northeastern University, Graduate Research Project, Boston, MA • Designed and executed an end-to-end CRISPR/Cas9 HPRT1 knock-out and knock-in workflow in CHO cells, from guide selection and Cas9-RNP assembly through CRISPRmax lipofection, 6-thioguanine selection, and sequence-level genotyping. • Designed and ran an 8-point 3-fold CellTiter-Glo 2.0 dose-response in duplicate to set the 6-thioguanine selection window; diagnosed the null result as an unqualified readout rather than a biological effect, and defined the blank, full-kill and $Z' \geq 0.5$ system-suitability criteria required to qualify the assay. • Ran the genotyping train end to end (genomic DNA extraction, Phusion PCR, agarose gel electrophoresis, amplicon clean-up, Nanodrop QC, bidirectional Sanger sequencing with ICE indel deconvolution) and set A260/280 and A260/230 acceptance criteria after tracing read failures to template quantitation by absorbance. • Maintained CHO cultures across serial passage with trypan blue viability counts and growth characterization; ran puromycin and 6-thioguanine kill curves and restriction-digest identity checks on the donor knock-in construct.	Sept 2025 – Dec 2025
Reagene Biosciences Pvt Ltd, Research Intern, Bangalore, India • Built market research and competitive analysis for a 3D-bioprinted human tissue platform positioned as an alternative to animal testing in preclinical safety work. • Contributed to investor decks and grant proposals used in fundraising, translating the underlying science into a commercial case for non-technical funders. • Identified operational and delivery bottlenecks affecting customer fulfilment and raised process gaps with company leadership. • Maintained and cryopreserved mammalian cell lines at high viability to supply the bioink formulation process for 3D-bioprinted tissue models. • Ran molecular biology and recombinant protein characterization workflows supporting formulation development, and authored SOPs for routine laboratory procedures.	Aug 2023 – Aug 2024
Tranalab Pvt Ltd, Center for Human Genetics, Research Intern Bachelor's Thesis, Bangalore, India • Designed and executed a complete expression and purification process for M4, a recombinant membrane protein domain, as a bachelor's thesis project ranked first in the cohort. • Identified solubilization as the yield-limiting step of the process and reformulated the buffer accordingly, increasing soluble protein recovery by roughly 48%. • Ran the upstream and downstream train end to end: expression in <i>E. coli</i> under IPTG induction, cell lysis by probe sonication, inclusion body solubilization, and Ni-NTA affinity chromatography (IMAC) with stepwise wash and elution. • Confirmed product identity by SDS-PAGE, quantified yield by A280, buffer-exchanged by dialysis, and assessed refolding by affinity chromatography to verify retained binding function.	Jan 2023 – Aug 2023

Leadership and Projects

Process Development Report: CRISPR/Cas9 Knockout of HPRT1 in CHO Cells, Independent • Rewrote a completed CRISPR workflow as a process development document: unit operation breakdown with in-process controls, quality target profile with CQA and CPP criticality assessment, eight-item deviation and CAPA analysis, an ICH Q2 fit-for-purpose qualification plan for the viability readout, and a resolution IV fractional factorial screen over six delivery parameters with a face-centered central composite follow-on.	2026
Vice President, Northeastern University, Graduate Biotech-Bioinfo Association, Boston, MA • Led executive board operations and strategy, managing communications and cross-committee coordination, and planned networking events, workshops, and industry panel discussions.	Dec 2024 – Jan 2026
CurcuNano, Chief Executive Officer, BIOT 5219 Biotech Enterprise, Northeastern University • Led the venture team as CEO for a curcumin nanoformulation concept, developing the business case and market positioning, and owned the funding ask in the final investor pitch, covering raise amount, use of funds and investment rationale, defended to a faculty panel.	Spring 2026